Anatomical Structure versus Generative Fidelity: A Hierarchical VAE Study of Chest Radiographs

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Abstract

We evaluate a from-scratch hierarchical VAE (H3B) and decoder-only adversarial refinement (H6A) for 128×128 pediatric chest radiographs. Combining a 128-D global latent with a conditional $8 \times 32 \times 32$ spatial latent, H3B reaches validation MSE 0.002467 and SSIM 0.7542 versus $0.003621/0.6898$ for an MSE baseline. H6A decreases matched-protocol FID from 216.52 to 210.14 (522 prior samples), but fine pulmonary detail remains unrealistic. We explicitly separate posterior reconstruction from unconditional generation.

1 Motivation and probabilistic model

VAEs enable amortized inference and ancestral sampling [Kingma and Welling, 2014, Rezende et al., 2014]; latent hierarchies support multi-scale structure [Maaløe et al., 2019, Vahdat and Kautz, 2020]. For chest radiographs, accurate lung silhouettes can coexist with blurred ribs and pulmonary markings. We test local-latent design and limited decoder refinement, using separate reconstruction and synthesis evaluations.

Let $x \in [0,1]^{1\times128\times128}$. The H3B generative factorization and amortized posterior are

$$\begin{aligned} p_\theta(x, z_g, z_\ell) &= p(z_g)\, p_\theta(z_\ell \mid z_g)\, p_\theta(x \mid z_g, z_\ell), \\ q_\phi(z_g, z_\ell \mid x) &= q_\phi(z_g \mid x)\, q_\phi(z_\ell \mid x, z_g), \qquad p(z_g) = \mathcal{N}(0, I). \end{aligned} \tag{1}$$

Both Gaussian posteriors use reparameterization; the local posterior predicts corrections relative to the conditional prior. There are no deterministic encoder–decoder skips. At generation time $z_g \sim \mathcal{N}(0, I)$, $z_\ell \sim p_\theta(z_\ell \mid z_g)$, with no input image.

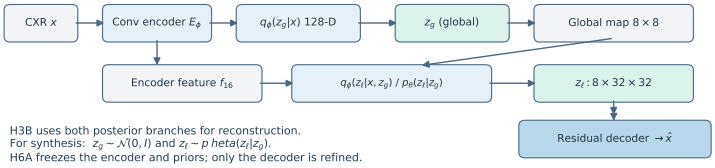


Figure 1: H3B: global and conditional spatial latent branches. H6A refines only the decoder.

Training objective. H3B minimizes the hybrid objective

$$\mathcal{L}_{\text{H3B}} = \underbrace{\|x - \hat{x}\|^2_{2,\text{sum}} + 48(1 - \text{SSIM}(x, \hat{x})) + 32\|S(x) - S(\hat{x})\|_1}_{\mathcal{L}_{\text{rec}}} + \beta_t \left[\text{KL}_g + 0.35\, \text{KL}_\ell\right], \tag{2}$$

where S is Sobel filtering, β_t warms up over 10 epochs, and SSIM measures structural fidelity [Wang et al., 2004]. The auxiliary penalties mean this is *not* an exact negative ELBO. Training: AdamW (1.5×10^{-4}, weight decay 10^{-4}), batch 16, 50 epochs, gradient clipping 1, EMA 0.999; select by validation objective.

Data protocol. The assigned pediatric CXR dataset [Kermany et al., 2018] contributes 4,694 training and 522 stratified validation images (seed 2026), resized to 128^2 grayscale. We do not evaluate on the supplied test folder.

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Table 1: Held-out posterior reconstruction; rows compare jointly changed model/loss settings.

Variant	Principal change	MSE↓	SSIM↑
H1	MSE-only baseline	0.003621	0.6898
H2	SSIM/Sobel + 16^2 local latent	0.003520	0.6971
H3A	32^2 local latent, default KL	0.003778	0.6926
H3B	32^2 local latent, $0.35\mathrm{KL}_\ell$	**0.002467**	0.7542
H3C	H3B + high-pass loss	0.002469	**0.7547**

2 Decoder refinement and evaluation

H6A: controlled optional refinement. H6A freezes H3B's encoder, global mapping, and local prior, updating only the decoder for 15 epochs ($\eta_G = 10^{-5}$; EMA 0.995). A spectral-normalized patch critic [Miyato et al., 2018] uses a hinge loss on real images, posterior reconstructions and independent prior samples. The decoder minimizes $\mathcal{L}_{\mathrm{rec}} + \lambda_t \mathcal{L}_{\mathrm{adv}}$, where $\mathcal{L}_{\mathrm{adv}} = -\frac{1}{2}[\mathbb{E}D(\hat{x}_{\mathrm{post}}) + \mathbb{E}D(\hat{x}_{\mathrm{prior}})]$. Adaptive gradient balancing targets a bounded $\sim 5\%$ adversarial/reconstruction ratio at the output layer. This is a VAE-derived adversarial refinement, not pure ELBO training [Larsen et al., 2016].

Matched distributional comparison. Matched evaluation uses seed 2026 and 522 independent prior draws against 522 validation images. FID/KID/IS [Heusel et al., 2017, Bińkowski et al., 2018, Salimans et al., 2016] consistently favor H6A (Table 2). The H3B reference is recomputed in this pipeline; its separate notebook FID 213.15 is not mixed with the paired estimate. Single-seed differences are descriptive.

Table 2: Matched independent synthesis (seed 2026; 522 real and 522 generated images).

Generator	FID↓	KID↓	IS↑
H3B (original HVAE)	216.52	0.2725	2.287
H6A (decoder refinement)	**210.14**	**0.2612**	**2.321**

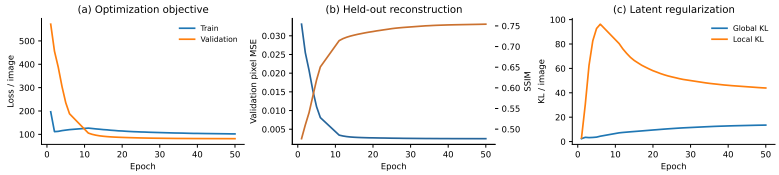


Figure 2: H3B training (50 epochs): objective, held-out MSE/SSIM, and global/local KL.

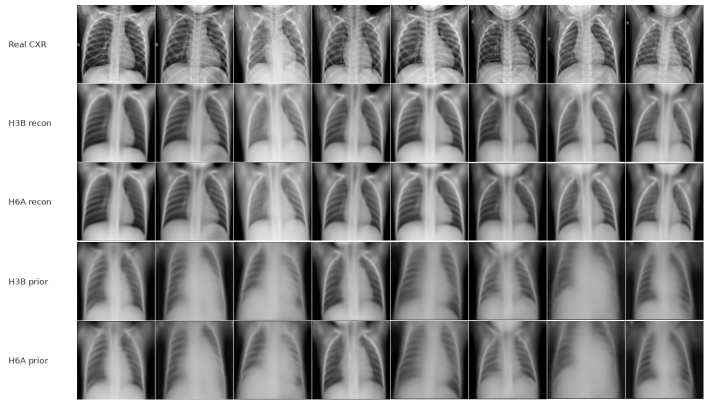


Figure 3: **Top:** real/H3B/H6A reconstructions (eight cases). **Bottom:** paired H3B/H6A independent prior samples (eight draws). The blocks are not image-to-image pairs.

Interpretation and limits. H3B lowers posterior MSE by 31.9% versus H1, yet unconditional images remain oversmoothed. H6A improves FID by 6.38 (2.95%), but not perceptual realism. A higher-resolution latent sharpened reconstructions while degrading prior samples, illustrating the reconstruction–generation gap. With only 522 images and ImageNet features, FID does not certify radiological validity [Woodland et al., 2024]. **Conclusion:** learned anatomical structure remains texture-deficient and not clinically validated.

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